

How these sheets work

One sheet per competency, 5 sheets for this video. Each gives you two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before the video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing, some a grid, some three panels. The division is yours.
4. Words go inside the drawing, as labels and inside little boxes with arrows. A sentence running the width of the box does not count.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a grid or a sequence, so whichever way you think, there is a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set it up on a tablet, and write the prompt letter you chose at the top. Hand ruled counts exactly the same.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. Function belongs to physiology and is not on these sheets.

The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

11

WEEK 1 · CELL ANATOMY

Generalized cell regions

COMP

You should be able to: Name the three basic regions of a generalized cell, the plasma membrane, cytoplasm, and nucleus, and state what each contains.

PICK ONE AND DO THAT ONE

☐ **A LABELED CELL**

Draw a generalized cell large. Label the plasma membrane, the cytoplasm, and the nucleus, then draw and label what each region contains: the membrane's layers, the cytosol and organelles, the nucleoplasm and chromatin.

Then: Shade the three regions three different ways so the borders are obvious.

☐ **B THREE-BOX HUB**

Draw three boxes, plasma membrane, cytoplasm, nucleus. Inside each, draw the things that belong to it. Draw arrows between the boxes for anything that crosses from one region into another.

Then: Write beside each box what it does for the cell, in a few words.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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12

WEEK 1 · CELL ANATOMY

Plasma membrane structure

COMP

You should be able to: Describe the plasma membrane components, distinguish integral from peripheral proteins, and identify the surface specializations microvilli, cilia, and flagellum.

PICK ONE AND DO THAT ONE

☐ **A MEMBRANE IN CROSS SECTION**

Draw a slice of plasma membrane large: the phospholipid bilayer with heads and tails, cholesterol, an integral protein crossing it, a peripheral protein sitting on it, and a carbohydrate on the outer face. Label every part.

Then: Above the membrane, draw microvilli, a cilium, and a flagellum side by side at the right relative sizes and label them.

☐ **B SORTING TABLE**

Draw a table with every membrane component down the side. Across the top: where in the bilayer it sits, what it looks like, and what it is for. Draw the component in the second column.

Then: Add three rows for microvilli, cilia, and flagellum, with the cell type you would find each on.

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13

WEEK 1 · CELL ANATOMY

Nucleus structure

COMP

You should be able to: Identify the nuclear envelope, nuclear pores, nucleolus, nucleoplasm, chromatin, and chromosomes and state the role of each.

PICK ONE AND DO THAT ONE

☐ **A LABELED NUCLEUS**

Draw a nucleus large. Draw the envelope as two membranes with pores through both, the nucleolus, the nucleoplasm, chromatin as loose threads, and one chromosome as the same thread coiled tight. Label all six.

Then: Draw an arrow through a pore and label what travels through it in each direction.

☐ **B INSIDE-OUT LIST WITH SKETCHES**

Draw the nucleus as six layers from outside in, each as its own small drawing: envelope, pores, nucleoplasm, chromatin, chromosome, nucleolus. Beside each, one phrase for what it is.

Then: Circle the two drawings that are the same material in two states.

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PASS 2 COLOR _____

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14

WEEK 1 · CELL ANATOMY

Organelle identification

COMP

You should be able to: Identify the membranous and non-membranous organelles and state the structural job of each, including the endomembrane system path from rough ER to exocytosis.

PICK ONE AND DO THAT ONE

☐ **A LABELED CELL, EVERY ORGANELLE**

Draw a cell and draw every organelle in it: rough and smooth ER, Golgi, lysosome, peroxisome, mitochondrion, ribosomes, centrioles, cytoskeleton. Label each and mark whether it has a membrane.

Then: Draw the path of a protein from rough ER to Golgi to vesicle to the membrane and out, as numbered arrows.

☐ **B TWO-COLUMN SORT**

Draw two columns headed membranous and non-membranous. Draw each organelle in the right column with its name and a few words for its job.

Then: Under the table, draw the endomembrane path as a chain of boxes with arrows, rough ER first.

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PASS 2 COLOR _____

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15

WEEK 1 · TISSUES AND HISTOLOGY

Cell junctions

COMP

You should be able to: Identify the five epithelial cell junctions, what each does, what it links to, and its major protein.

PICK ONE AND DO THAT ONE

- ☐ **A TWO CELLS SIDE BY SIDE**
Draw two tall epithelial cells side by side on a basement membrane. Draw every junction between them and beneath them where it belongs, top to bottom, and label each with its name and its protein.

Then: Draw an arrow showing what each junction stops or lets through.
- ☐ **B JUNCTION TABLE**
Draw a table with the five junctions down the side. Across the top: what it does, what it links to, its main protein, and a tiny sketch of its shape.

Then: Mark the junction that links cell to basement membrane rather than cell to cell.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

split it up however the prompt needs